Bachelor of Computer

Applications Semester – V

Business Analytics

Experiment No.: 3

Title: Building a Dashboard

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Objective

The objective of this experiment is to gain hands-on experience in importing business data from multiple heterogeneous sources such as Excel files, CSV files and databases into Microsoft Power BI. The experiment focuses on establishing connections between different datasets, cleaning and transforming the data using Power Query, building a proper data model with relationships between tables, and finally designing an interactive dashboard that communicates key business metrics clearly to stakeholders. This exercise simulates a real-world scenario faced by data analysts in a retail organisation where sales, customer and product information originate from different systems and must be consolidated into a single source of truth for reporting.

Problem Statement

As a Data Analyst in a retail organisation, sales, customer and product data are maintained in separate sources such as Excel files, CSV files and other databases. Management requires a centralised reporting system to monitor business performance. The objective of this experiment is to import data from multiple sources into Power BI, establish the required relationships between the datasets, transform the data for analysis, and build an interactive dashboard that presents key business metrics such as total sales, order quantity, customer distribution and product performance.

Without a centralised reporting mechanism, management is forced to manually consolidate reports from different departments, which is time-consuming and prone to error. A well-designed Power BI dashboard eliminates this manual effort by automatically refreshing data from the connected sources and presenting it through visually intuitive charts, KPI cards and slicers, allowing decision-makers to identify trends and anomalies at a glance.

Tools and Technologies Used

Tool / Technology	Purpose
Microsoft Power BI Desktop	Data import, transformation, modeling and dashboard design
Power Query Editor	Data cleaning and transformation (ETL)
DAX (Data Analysis Expressions)	Creating calculated measures and columns
Microsoft Excel / CSV	Source files for sales, customer and product data
MySQL (optional source)	Relational database connection for extended datasets

Dataset Description

The dataset used for this experiment consists of order-level transaction records along with associated customer and product information. The key fields used across the report are described below.

Field Name	Description
OrderDate	Date on which the order was placed
OrderQuantity	Number of units ordered in a transaction
ProductPrice	Unit price of the product sold
ProductName	Name of the product purchased (e.g., AWC Logo Cap, Water Bottle)
CustomerKey	Unique identifier for each customer
Gender	Gender of the customer (M / F / NA)
Total Sales	Calculated measure: OrderQuantity × ProductPrice

Methodology / Steps Followed

Step 1: Data Collection

The first step involved identifying and gathering the raw data required for analysis. Data was sourced from multiple formats to replicate a realistic business environment where information rarely resides in a single system.

- Collected order and sales data containing fields such as OrderDate, OrderQuantity and ProductPrice.
- Collected customer data including CustomerKey and Gender.
- Collected product data including ProductName and related attributes.

Step 2: Data Import

Once the source files were identified, they were connected to Power BI using the built-in connectors, ensuring each table loaded correctly with the right data types.

- Opened Power BI Desktop and used the Get Data option to connect to the source files.
- Selected the appropriate connector (Excel Workbook / Text-CSV / Database) for each source.
- Imported the sales, customer and product datasets into Power BI.
- Loaded all tables into the Power BI data model.

Step 3: Data Cleaning and Transformation

Raw data is rarely analysis-ready. The Power Query Editor was used to profile and clean each table before loading it into the model.

- Removed duplicate and blank records using Power Query Editor.
- Handled missing values in key columns.
- Renamed columns for clarity (e.g., OrderQuantity, ProductPrice, CustomerKey).
- Converted data types (dates, numeric fields) to appropriate formats.
- Filtered out irrelevant or test records that did not belong to the reporting period.

Step 4: Data Modeling

After cleaning, the individual tables were linked together to form a coherent data model, following a star-schema approach with the Sales table at the centre and Customer and Product as dimension tables.

- Created relationships between the Sales, Customer and Product tables.
- Verified one-to-many relationships using the Model view.
- Created measures such as Total Sales, Total Orders, Total Quantity and Total Customers using DAX.

Key DAX Measures

Measure	DAX Formula (Simplified)	Description
Total Sales	SUM(Sales[OrderQuantity] * Sales[ProductPrice])	Total revenue generated
Total Orders	DISTINCTCOUNT(Sales[OrderID])	Count of unique orders placed
Total Quantity	SUM(Sales[OrderQuantity])	Total units sold
Total Customers	DISTINCTCOUNT(Sales[CustomerKey])	Count of unique customers

Step 5: Dashboard Development

With the model and measures in place, two dashboard views were designed to present the data from complementary angles — one focused on order quantity patterns and the other on overall sales performance.

- Created KPI cards for Total Sales, Total Orders, Total Quantity and Total Customers.
- Developed a line chart for Sum of OrderQuantity / Total Sales by OrderDate to visualise monthly trends.
- Created bar charts for OrderQuantity and Total Sales by ProductName.
- Built a scatter chart of OrderQuantity against ProductPrice.
- Added a donut chart showing OrderQuantity distribution by Gender.
- Created a horizontal bar chart of OrderQuantity by CustomerKey.
- Designed an interactive, easy-to-read business dashboard using a consistent colour theme.

Output

The dashboards developed in Power BI are shown below.

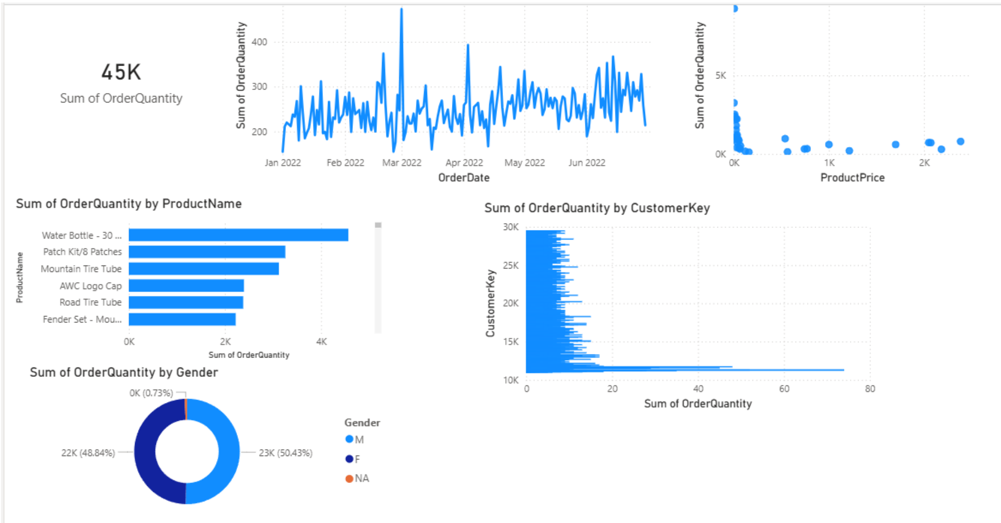

Fig 1: Order Quantity Analysis Dashboard

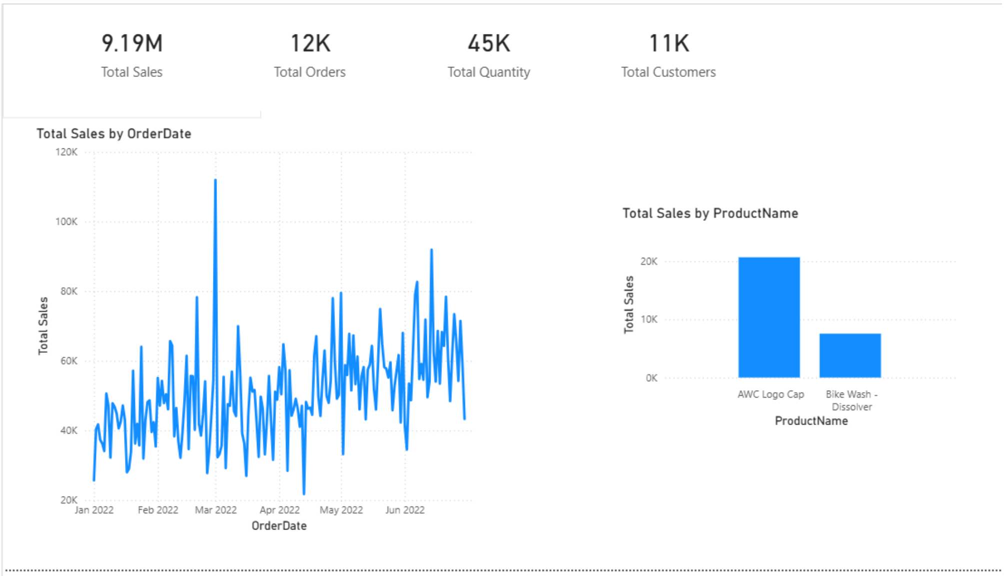

Fig 2: Sales Overview Dashboard

The final Power BI dashboards provide:

- Total Sales, Total Orders, Total Quantity and Total Customers at a glance.
- Order Quantity trend by OrderDate.
- Product-wise Sales and Order Quantity Analysis.
- Customer distribution by Gender and CustomerKey.
- Relationship between Order Quantity and Product Price.

The dashboards enable management to monitor overall business performance, identify top-selling products, and understand customer demographics, thereby supporting data-driven decision-making.

Dashboard Insights and Analysis

Analysis of the Order Quantity Analysis Dashboard (Fig 1) shows that the total order quantity across the reporting period stands at 45K units. The line chart of OrderQuantity by OrderDate reveals a generally stable order pattern between January and June 2022, with a sharp spike around March 2022 indicating a possible seasonal promotion or bulk order event. The scatter plot of OrderQuantity against ProductPrice indicates that lower-priced items are ordered far more frequently than higher-priced items, which is typical of fast-moving consumer accessories.

The bar chart of OrderQuantity by ProductName shows that the Water Bottle - 30 oz, Patch Kit/8 Patches and Mountain Tire Tube are the top-selling products by volume, suggesting these are low-cost, high-frequency purchase items. The donut chart shows an almost even split between Male (48.84%) and Female (50.43%) customers, with a negligible unspecified (NA) segment, indicating a balanced customer base across genders. The horizontal bar chart of OrderQuantity by CustomerKey highlights that a small group of customers accounts for a disproportionately large share of total order volume.

The Sales Overview Dashboard (Fig 2) summarises overall performance through four KPI cards: Total Sales of 9.19M, Total Orders of 12K, Total Quantity of 45K and Total Customers of 11K. The Total Sales by OrderDate trend line mirrors the order quantity pattern, confirming the March 2022 spike, along with a gradual upward trend from April to June 2022. The Total Sales by ProductName chart shows that the AWC Logo Cap contributes significantly more revenue than the Bike Wash - Dissolver, even though both are lower-priced items, indicating stronger overall demand for the Logo Cap.

Together, both dashboards give management a 360-degree view: the first focuses on volume and customer-level patterns, while the second focuses on revenue and product-level performance, allowing decisions on inventory planning, targeted marketing and customer retention to be made with confidence.

Challenges Faced

- Handling inconsistent date formats between the Excel and CSV sources during import.
- Identifying and removing duplicate customer records without losing valid transactions.
- Optimising DAX measures so that dashboard visuals refreshed quickly on a large order dataset.
- Choosing chart types that clearly convey both volume (OrderQuantity) and value (Sales) without cluttering the dashboard.

Conclusion

This experiment successfully demonstrated the process of importing data from multiple sources into Power BI and transforming it into meaningful business insights. Data cleaning, transformation and modeling techniques were applied to ensure accurate reporting. The developed dashboards provide a centralised view of sales, customer and product performance and support better business decision-making.

The exercise reinforced the importance of a well-structured data model and clean data as the foundation of any reliable business intelligence solution. By combining KPI cards, trend lines, bar charts, scatter plots and donut charts, the resulting dashboards translate raw transactional data into actionable insights that a non-technical business audience can interpret at a glance.

Learning Outcomes

Through this experiment, I learned:

- How to import data from Excel and CSV files into Power BI.
- Data cleaning and transformation using Power Query.
- Creating relationships between multiple tables.
- Developing DAX measures and calculated columns.
- Designing interactive dashboards with KPI cards, charts and slicers.
- Interpreting visualised data to draw meaningful business conclusions.
- Generating business insights from raw data.

Project Repository

The Power BI file (.pbix), source datasets and supporting documentation for this experiment have been uploaded to the following GitHub repository for reference and version control:

GitHub Link: https://github.com/AarushCS777/Aarush_BusinessAnalytics